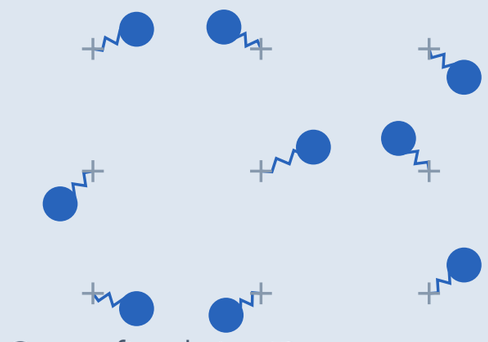
Einstein crystal

One solvable model connects atomic vibrations, quantum heat capacity, entropy, and solid free energies.

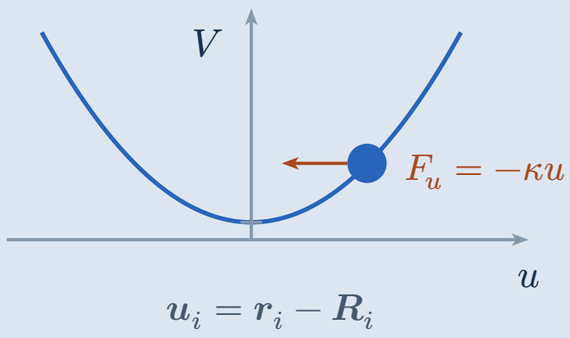
1 Replace the crystal by local springs

N atoms × 3 directions = M = 3N independent modes



Cross: fixed site $\boldsymbol{R}_i$
Dot: atom at $\boldsymbol{r}_i$

One displacement coordinate



$$\boldsymbol{u}_i = \boldsymbol{r}_i - \boldsymbol{R}_i$$

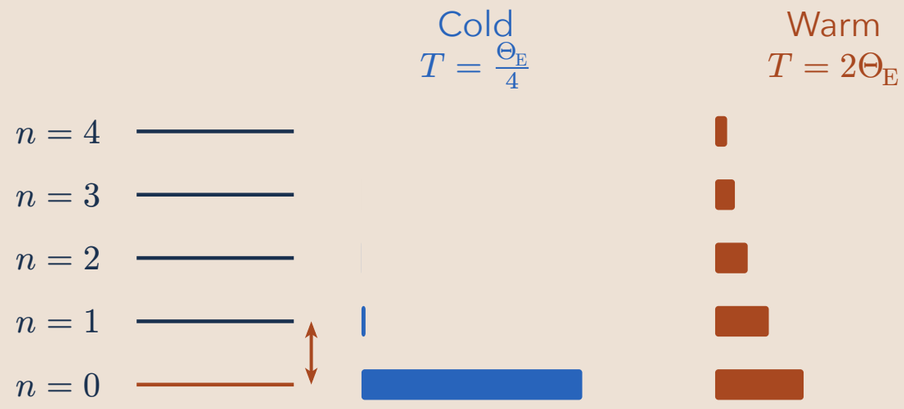
$$V_{\mathrm{E}} = \frac{\kappa}{2}\sum_{i=1}^{N}|\boldsymbol{u}_i|^2 \quad \omega_{\mathrm{E}} = \sqrt{\frac{\kappa}{m}}$$

Same stiffness κ and mass m; springs attach to sites, not neighboring atoms.

Read the restoring force
$\boldsymbol{u}_i$ is atom i's displacement from its fixed site. The force points back toward the potential minimum: $F_u = -\kappa u$. Doubling $|u|$ quadruples the spring energy. Stiffer springs or lighter atoms have a higher angular frequency ω_{E}. This harmonic approximation describes small vibrations about stable sites.

2 Heating fills a quantum ladder

$$E_n = \left(n + \tfrac{1}{2}\right)\hbar\omega_{\mathrm{E}} \quad n = 0, 1, 2, \ldots$$



$P(n) = (1 - e^{-x})e^{-nx}$; bar length = probability.

Gap $\hbar\omega_{\mathrm{E}}$; the ground state still has $E_0 = \hbar\frac{\omega_{\mathrm{E}}}{2}$. Higher levels omitted.

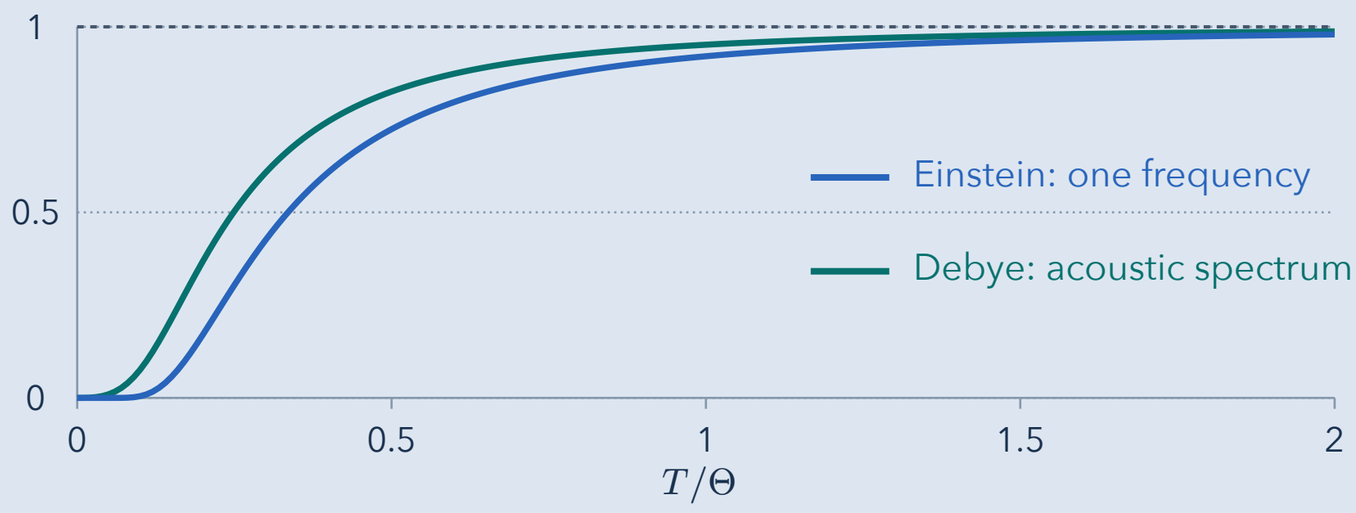
Thermal energy competes with the gap
n counts excitation quanta of one mode; $P(n)$ is its level probability. Θ_{E} expresses the energy gap as a temperature. Thus $x = \frac{\Theta_{\mathrm{E}}}{T}$ compares the gap $\hbar\omega_{\mathrm{E}}$ with thermal energy $k_{\mathrm{B}}T$: large x means few excited states.

3 Why the heat capacity falls on cooling

Thermal excitations store heat. The fixed zero-point energy contributes no heat capacity.

$$\frac{C_V}{3Nk_{\mathrm{B}}} = \frac{x^2 e^x}{(e^x - 1)^2} \quad x = \frac{\Theta_{\mathrm{E}}}{T} \quad \Theta_{\mathrm{E}} = \frac{\hbar\omega_{\mathrm{E}}}{k_{\mathrm{B}}}$$

Heat capacity / $3Nk_{\mathrm{B}}$ Classical Dulong–Petit limit: 1



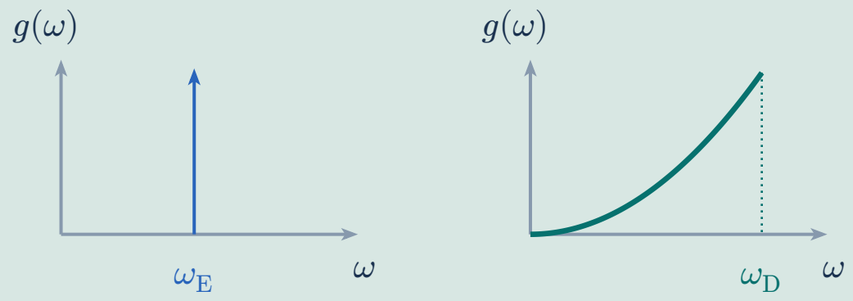
Cold Einstein: $C_V \propto x^2 e^{-x}$ · Hot: $C_V \to 3Nk_{\mathrm{B}}$
Equal $\Theta_{\mathrm{E}} = \Theta_{\mathrm{D}} = \Theta$ for shape comparison; real fitted scales differ.

Heat capacity measures energy stored per degree
$C_V = \left(\frac{\partial U}{\partial T}\right)_V$ measures how much energy a small temperature rise stores at fixed volume. Classically, each mode contributes $\frac{k_{\mathrm{B}}}{2}$ from motion and $\frac{k_{\mathrm{B}}}{2}$ from its spring, so $3N$ modes give $3Nk_{\mathrm{B}}$. At low T, a mode stores little extra heat when its excitation gap greatly exceeds $k_{\mathrm{B}}T$.

4 Real crystals have phonons



Coupled atoms → many normal-mode frequencies



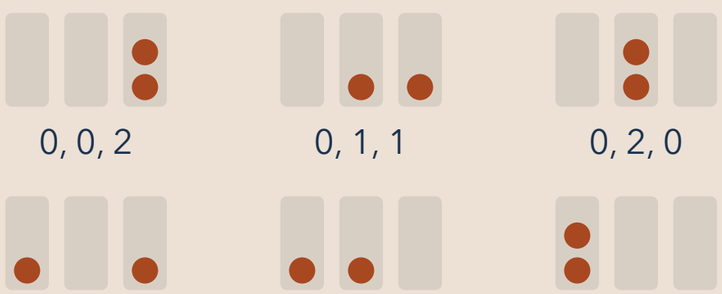
One frequency — Debye: $g(\omega) \propto \omega^2$

Each spectrum contains 3N modes. In 3D, low-frequency acoustic modes give $C_V \propto T^3$.
Harmonic normal modes remain independent.

Why soft modes matter
$g(\omega)\,\mathrm{d}\omega$ counts modes in a frequency interval. Soft acoustic modes remain easy to excite as the solid cools, giving $C_V \propto T^3$ at low T. ω_{D} sets the cutoff, with $\Theta_{\mathrm{D}} = \hbar\frac{\omega_{\mathrm{D}}}{k_{\mathrm{B}}}$.

5 Count states to get entropy

One atom: M = 3 modes, q = 2 quanta.
Six allocations have the same energy.



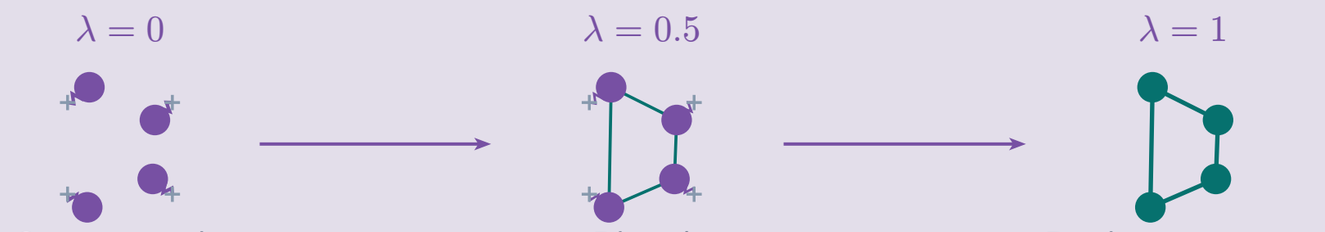
$$\Omega(M, q) = \binom{q+M-1}{q}$$

$$S = k_{\mathrm{B}} \ln \Omega \quad M = 3N$$

Same energy, different microstates
Each bin is an x, y, or z mode. The total excitation energy is $q\hbar\omega_{\mathrm{E}}$ above the ground state. Ω counts allocations at fixed q: more possibilities mean larger entropy. Here $\Omega = 6$ and $S = k_{\mathrm{B}} \ln 6$.

6 A reference for solid free energies

Classical Frenkel–Ladd integration · Helmholtz free energy $F = U - TS$



$$V_\lambda = (1 - \lambda)V_{\mathrm{E}} + \lambda V_{\mathrm{real}}$$

$$F_{\mathrm{real}} - F_{\mathrm{E}} = \int_0^1 \langle V_{\mathrm{real}} - V_{\mathrm{E}} \rangle_\lambda \,\mathrm{d}\lambda$$

$\langle \ldots \rangle_\lambda$: equilibrium average in the blend. Keep temperature, volume, and the positional constraint fixed along a reversible crystal path. Release the fixed center of mass (or one particle: Einstein molecule) with its analytic free-energy correction.

Use: compare candidate crystal structures
$\lambda \in [0, 1]$ controls the blend. Sampling each value gives the average in the integral, which adds up the reversible free-energy change. The result includes both energy and entropy through $F = U - TS$. At the same temperature, volume, and particle number, the candidate with lower Helmholtz free energy F is favored.

Symbols: T temperature · k_{B} Boltzmann constant · $\hbar$ reduced Planck constant · C_V heat capacity at fixed volume · $g(\omega)$ mode density.

U internal energy · V potential energy · q excitation quanta. Heat curves include lattice vibrations only; no electronic or magnetic contribution. Thermal equilibrium is assumed.